# 연 규 환

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## EDUCATION

**한양대학교 자동차전자제어공학과 석사** Feb.2019

**한양대학교 미래자동차공학과 학사** Feb.2017

## RESEARCH INTERESTS

- Autonomous driving SW
- Deep-learning

## RESEARCH CAREER

**Researcher, Automotive Control & Electronics Lab, Hanyang University** (2017~2019)

- Ego-vehicle speed prediction (Deep-learning)
  - LSTM-RNN을 활용한 차량의 속도 예측
  - Nvidia-TX2 보드를 활용하여 딥러닝 모델 임베디드화, Real time performance 만족
  - 차량의 내부 통신 정보와 외부 정밀 GPS를 활용한 차량의 속도 예측 정확도 향상
- Intelligent Driver Model (Traffic flow modeling - Microscopic approach)
  - 운전자 성향을 반영한 파라미터 기반의 드라이버 모델링
  - 모델의 MCU 임베디드화

**Researcher, Powertrain Control, Hyundai Motor Company** (2019~Present)

- AUTOSAR based embedded system software engineer
  - Control system developer
  - Model-based design
  - Embedded system
  - ISO-26262, Software testing

## SKILLS

**Computer Languages:** C++, Python

**Embedded system:** Nvidia TX2, TC237 of Infineon

**Others:** ROS, CAN-communication

## PAPER

- Kyuhwan Yeon, Kyunghan Min, Jaewoork Shin, Myoungh Sunwoo, and Manbae Han, “EGO-VEHICLE SPEED PREDICTION USING A LONG SHORT-TERM MEMORY BASED RECURRENT NEURAL NETWORK”, International Journal of Automotive Technology, August 2019.
- Kyunghan Min, Kyuhwan Yeon, Yuhyeok Jo, Gyubin Sim, Myoungho Sunwoo, Manbae Han, “VEHICLE DECELERATION PREDICTION BASED ON DEEP NEURAL NETWORK AT BRAKING CONDITIONS”, International Journal of Automotive Technology, February 2020.

## CONFERENCE

- K. Min, K. Yeon, G. Sim, and M. Sunwoo “Driver characteristics based deceleration model for smart regenerative braking system”, Aachen Colloquium China 2018, Speaker

## PROJECT EXPERIENCE

### 1. Ego- vehicle speed prediction using a long-short term memory based RNNs

2018~2019.7

- Design of the ego-vehicle speed prediction model using long short-term memory (LSTM) based recurrent neural networks (RNNs)
  - Microscopic and non-parametric approach
  - Improvement of prediction accuracy on real urban roads by using various inputs
  - Internal vehicle information, relative distance and speed, location of the ego-vehicle
  - Implementation of the proposed model in the embedded system (Real driving test)
  - Language: Python, C, Keras, Tensorflow

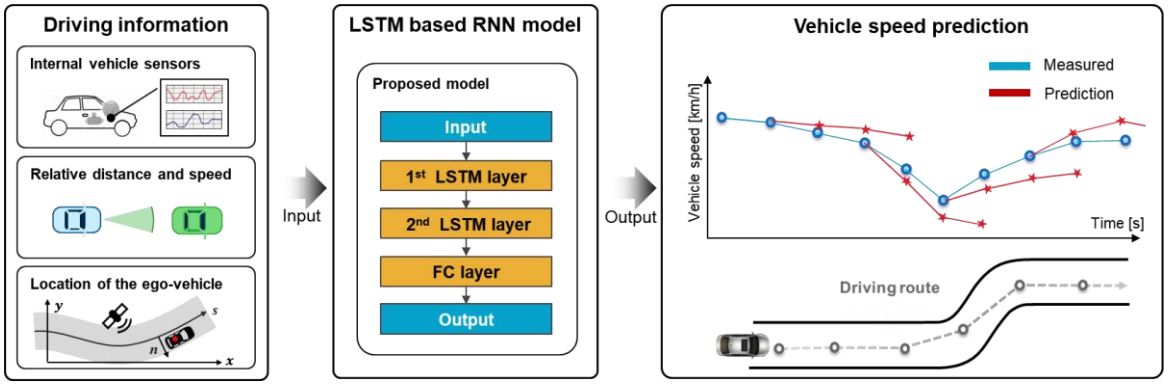



**Figure 1 Concept of the vehicle speed model**

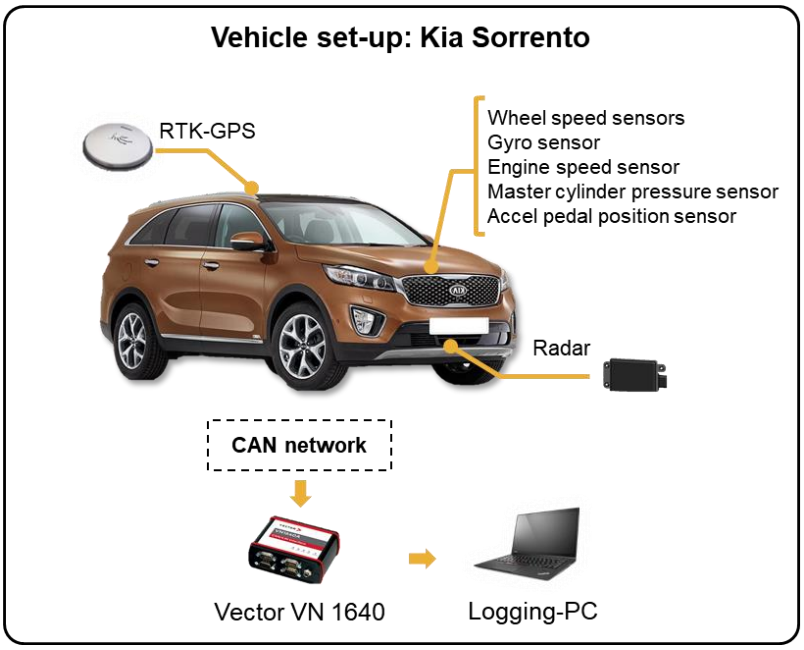



**Figure 2 Experiment environment**

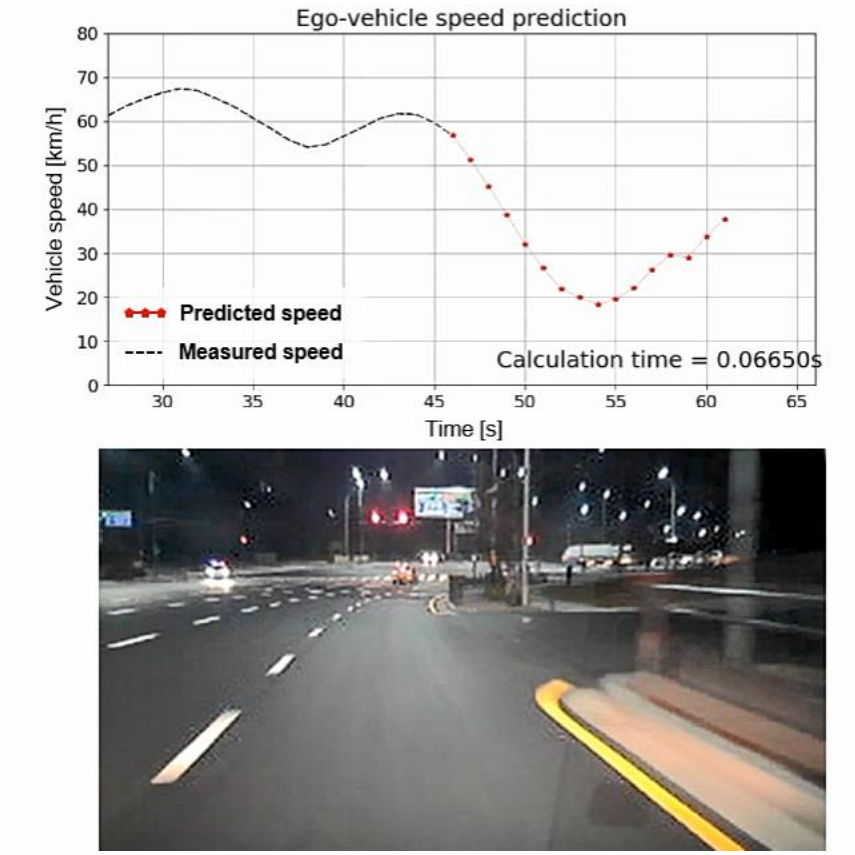
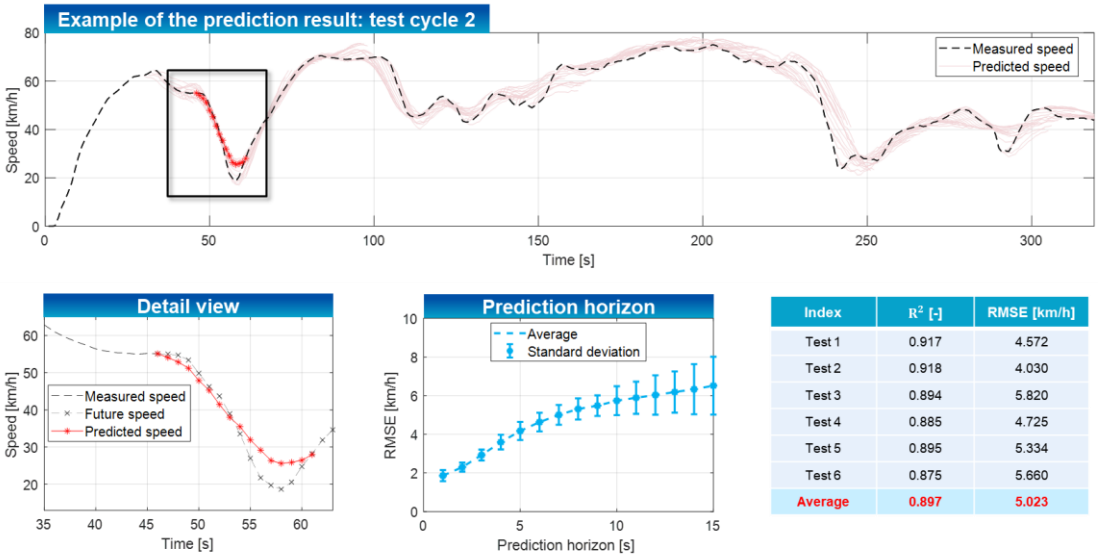



**Figure 3 Test result in real-urban driving (Real-time performance)**




| Index | $R^2$ [-] | RMSE [km/h] |
|---|---|---|
| Test 1 | 0.917 | 4.572 |
| Test 2 | 0.918 | 4.030 |
| Test 3 | 0.894 | 5.820 |
| Test 4 | 0.885 | 4.725 |
| Test 5 | 0.895 | 5.334 |
| Test 6 | 0.875 | 5.660 |
| Average | 0.897 | 5.023 |

**Figure 4 Evaluation of the proposed model along the prediction horizon**

- An ego-vehicle speed prediction model using LSTM based RNNs was proposed to improve the prediction performance in various driving conditions
  - The proposed model uses several model inputs
  - Internal vehicle sensors, a radar sensor, and the location information of the ego-vehicle
- **The proposed model was validated by real-driving data from a vehicle equipped with the radar and RTK-GPS**
  - The RMSE of test data was 5.023 km/h
  - The maximum calculation time was 0.073 seconds in the embedded system

## 2. 2019 Jeju BigData Competition - 퇴근시간 버스승차인원 예측 (Dacon) 2019.11~12

- 제주시 버스 정류장별 퇴근시간대의 버스 승차인원 모델링

  - Feature engineering (Bus-stop ride-take-off data, Traffic data, GIS 기반 관광객 Flow data, Weather …)
  - Data-driven modeling (Random-Forest , CNN, MLP)
  - https://dacon.io/cpt13/228543
  - Team leader, Data-preprocessing and modeling
  - Language and tool: Python, QGIS
  - RMSE in test set: 2.32262

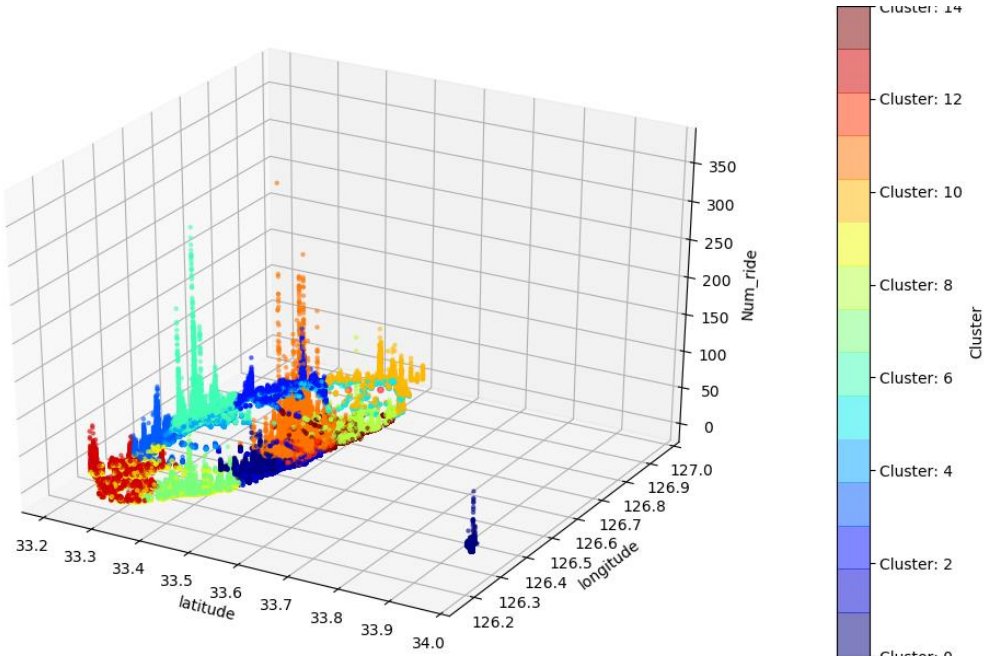



**Figure 5 Bus-stop ride data modeling**

## 3. Driver Characteristics based Deceleration Model 2018.10 ~11

- Design of the parametric deceleration model based on driver characteristics using intelligent driver model
  - Driver modeling from real-braking data
  - Analysis of the driver characteristics based on selected model parameters
  - Embedded in low-price micro-controller

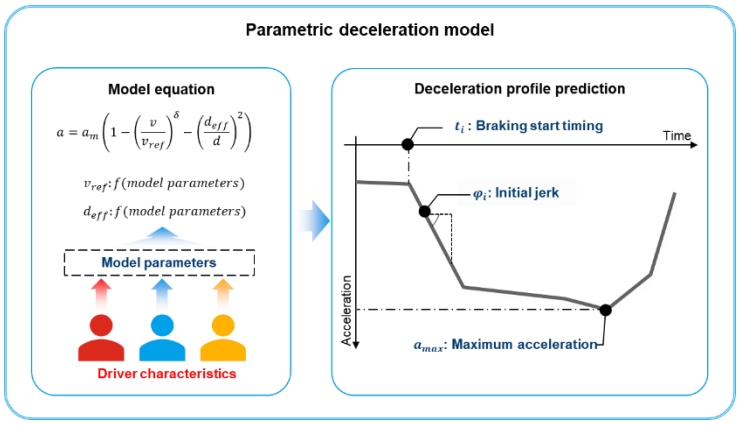



**Figure 6 Deceleration model concept**

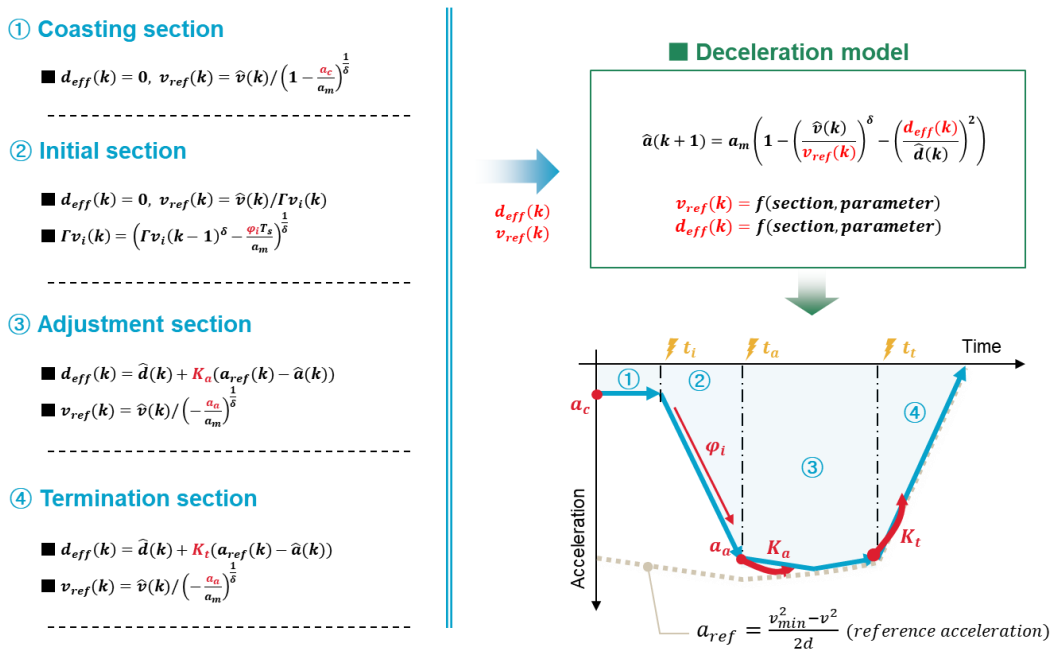



**Figure 7 Parameter update process**

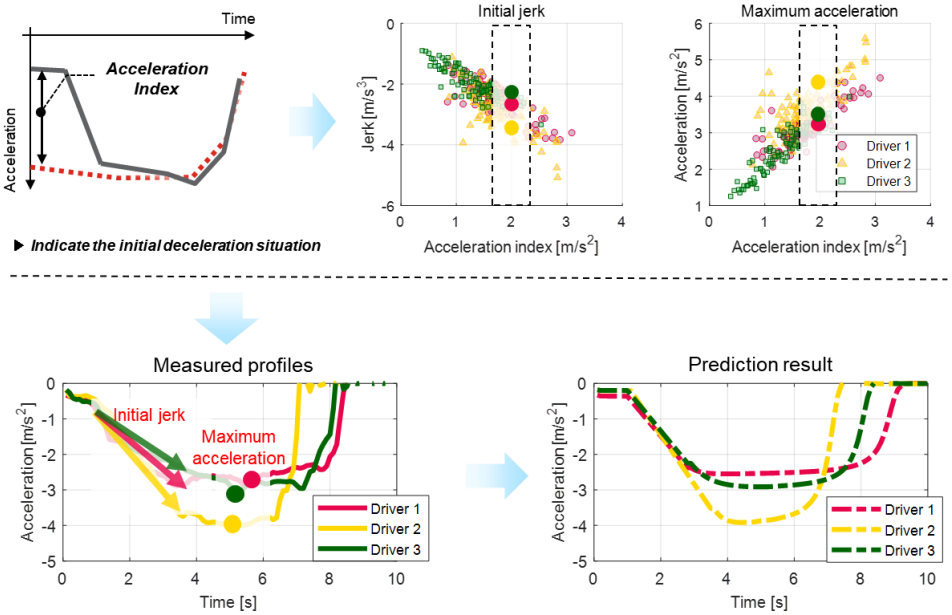



**Figure 8 Analysis of driver and modeling result**